

Coding Competency Test

1. Foo Bar

Question

Write a simple console program that prints the number from 1 to n, for each number x :

- print "foo", if x is divisible by 3
- print "bar", if x is divisible by 5
- print "foobar", if x is divisible by 3 and 5
- print the number itself, if x satisfies none of the rule Here's a sample output of such program with n=15

```
1, 2, foo, 4, bar, foo, 7, 8, foo, bar, 11, foo, 13, 14, foobar
```

Answer

We should loop for N,

and each loop we check the rules applied to the number.

to make sure all rules are checked, we do not use else in if statement

construct the output for each number by those rules.

then consolidate or it to final result

and print this final result

```
class Program
{
    static void Main(string[] args)
    {
        Console.WriteLine("Input N :");
        string input = Console.ReadLine();
        int number;
        if (!int.TryParse(input, out number))
        {
            Console.WriteLine("Invalid input. Please enter a valid integer.");
            return;
        }

        List<string> results = new List<string>();

        for (int x = 1; x <= number; x++)
        {
            string result = "";

            if (x % 3 == 0) result += "foo";
            if (x % 5 == 0) result += "bar";

            if (string.IsNullOrEmpty(result))
            {
                results.Add(x.ToString());
            }
        }
    }
}
```

```

        }
        else
        {
            results.Add(result);
        }
    }

    Console.WriteLine(string.Join(", ", results));
}
}

```

2. Foo Bar Jazz

Question

Continuing on the previous question. Add the following rules :

- print "jazz", if x is divisible by 7

This means for x=21, x=35 and x=105 the program should print "foojazz", "barjazz" and "foobarjazz" respectively.

Answer

Add new rule (jazz) to the validation section

```
if (x % 7 == 0) result += "jazz";
```

```

class Program
{
    static void Main(string[] args)
    {
        Console.WriteLine("Input N :");
        string input = Console.ReadLine();
        int number;
        if (!int.TryParse(input, out number))
        {
            Console.WriteLine("Invalid input. Please enter a valid integer.");
            return;
        }

        List<string> results = new List<string>();

        for (int x = 1; x <= number; x++)
        {
            string result = "";

            if (x % 3 == 0) result += "foo";
            if (x % 5 == 0) result += "bar";
            if (x % 7 == 0) result += "jazz";

            if (string.IsNullOrEmpty(result))

```

```
        {
            results.Add(x.ToString());
        }
        else
        {
            results.Add(result);
        }
    }

    Console.WriteLine(string.Join(", ", results));
}
}
```

3. Foo Baz Bar Jazz Huzz

Question

Continuing on the previous question. Using the same divisible logic, use the table below as the rules :

- 3 : "foo"
- 4 : "baz"
- 5 : "bar"
- 7 : "jazz"
- 9 : "huzz"

Answer

Add new rules to the validation section

```
if (x % 3 == 0) result += "foo";
if (x % 4 == 0) result += "baz";
if (x % 5 == 0) result += "bar";
if (x % 7 == 0) result += "jazz";
if (x % 9 == 0) result += "huzz";
```

so the final code will be

```
class Program
{
    static void Main(string[] args)
    {
        Console.WriteLine("Input N :");
        string input = Console.ReadLine();
        int number;
        if (!int.TryParse(input, out number))
        {
            Console.WriteLine("Invalid input. Please enter a valid integer.");
            return;
        }
    }
}
```

```
List<string> results = new List<string>();

for (int x = 1; x <= number; x++)
{
    string result = "";

    if (x % 3 == 0) result += "foo";
    if (x % 4 == 0) result += "baz";
    if (x % 5 == 0) result += "bar";
    if (x % 7 == 0) result += "jazz";
    if (x % 9 == 0) result += "huzz";

    if (string.IsNullOrEmpty(result))
    {
        results.Add(x.ToString());
    }
    else
    {
        results.Add(result);
    }
}

Console.WriteLine(string.Join(", ", results));
}
```